ES8156

# High Performance Stereo Audio DAC

## FEATURES

- High performance and low power multi-bit delta-sigma audio DAC
- 110 dB signal to noise ratio, -80 dB THD+N
- 24-bit, 8 to 96 kHz sampling frequency
- Integrated headphone driver with capless option
- Differential output for higher SNR and CMRR
- $I^2S$/PCM master or slave serial data port
- 256/384Fs, USB 12/24 MHz and other non standard audio system clocks
- $I^2C$ interface
- 7-band fully adjustable EQ
- Dynamic range compression
- Playback signal feedback
- Pop and click noise suppression
- 1.8V to 3.3V operation

## APPLICATIONS

- Headphone
- Speaker
- TV
- Portable audio devices

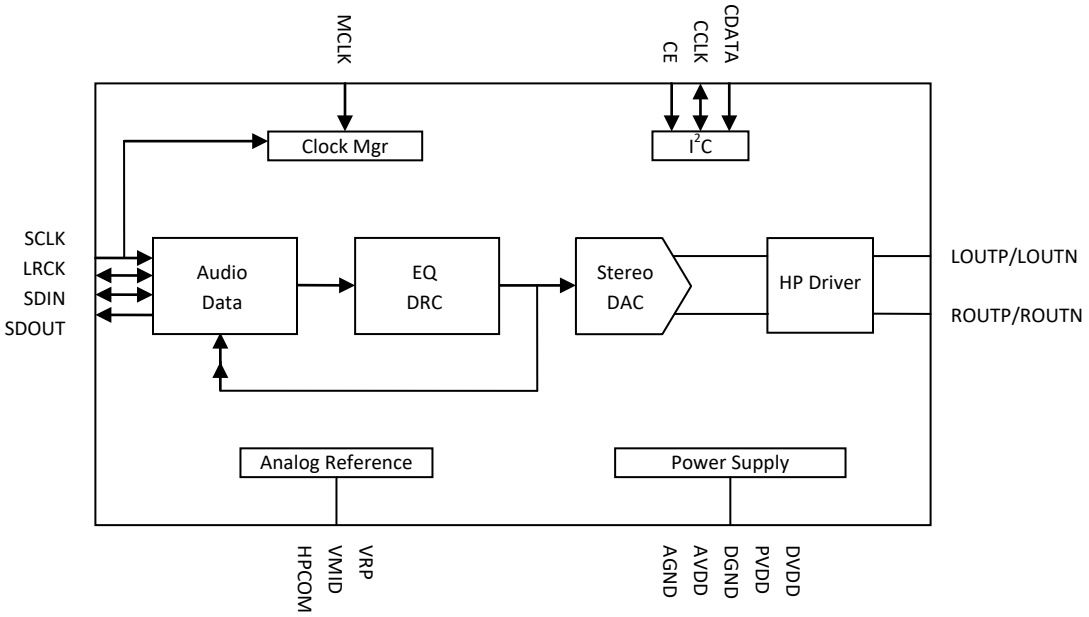
## ORDERING INFORMATION

ES8156 -40°C ~ +85°C
QFN-20

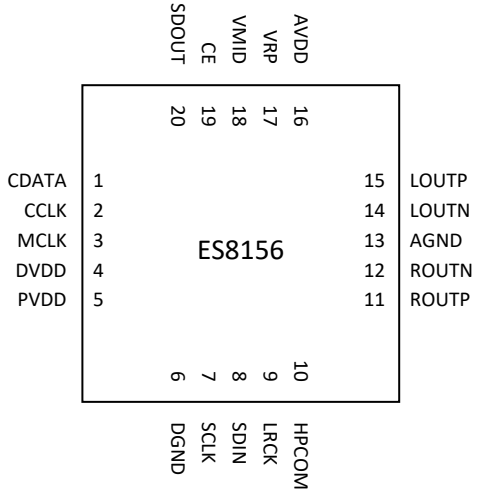
*1. BLOCK DIAGRAM ..... 4*
*2. PIN OUT AND DESCRIPTION ..... 5*
*3. TYPICAL APPLICATION CIRCUIT ..... 6*
*4. HARDWARE MODE ..... 8*
*5. CLOCK MODES AND SAMPLING FREQUENCIES ..... 8*
*6. MICRO-CONTROLLER CONFIGURATION INTERFACE ..... 8*
*7. DIGITAL AUDIO INTERFACE ..... 10*
*8. ELECTRICAL CHARACTERISTICS ..... 11*
ABSOLUTE MAXIMUM RATINGS ..... 11
RECOMMENDED OPERATING CONDITIONS ..... 11
DAC ANALOG AND FILTER CHARACTERISTICS AND SPECIFICATIONS ..... 11
DC CHARACTERISTICS ..... 12
$I^2C$ SWITCHING SPECIFICATIONS (SLOW SPEED MODE/HIGH SPEED MODE) ..... 12
SERIAL AUDIO PORT SWITCHING SPECIFICATIONS ..... 13
*9. CONFIGURATION REGISTER DEFINITION ..... 14*
REGISTER 0X00 – RESET CONTROL, DEFAULT 00011100 ..... 14
REGISTER 0X01 – MAIN CLOCK CONTROL, DEFAULT 00100000 ..... 14
REGISTER 0X02 – MODE CONFIG 1, DEFAULT 00000000 ..... 14
REGISTER 0X03 – MASTER LRCK DIVIDER 1, DEFAULT 00000001 ..... 15
REGISTER 0X04 – MASTER LRCK DIVIDER 0, DEFAULT 00000000 ..... 15
REGISTER 0X05 – MASTER CLOCK CONTROL, DEFAULT 00000100 ..... 15
REGISTER 0X06 – NFS CONFIG, DEFAULT 00010001 ..... 15
REGISTER 0X07 – MISC CONTROL 1, DEFAULT 00000000 ..... 15
REGISTER 0X08 – CLOCK OFF, DEFAULT 00000110 ..... 16
REGISTER 0X09 – MISC CONTROL 2, DEFAULT 00000000 ..... 16
REGISTER 0X0A – TIME CONTROL 1, DEFAULT 01010000 ..... 16
REGISTER 0X0B – TIME CONTROL 2, DEFAULT 01010000 ..... 17
REGISTER 0X0C – CHIP STATUS, DEFAULT 00000000 ..... 17
REGISTER 0X0D – P2S CONTRL, DEFAULT 00010000 ..... 17
REGISTER 0X10 – DAC COUNTER PARAMETER, DEFAULT 01000000 ..... 17
REGISTER 0X11 – SDP INTERFACE CONFIG 1, DEFAULT 00000000 ..... 17
REGISTER 0X12 – AUTOMUTE CONTROL, DEFAULT 00000100 ..... 18
REGISTER 0X13 – MUTE CONTROL, DEFAULT 00010001 ..... 18
REGISTER 0X14 – VOLUME CONTROL, DEFAULT 10111111 ..... 19

REGISTER 0X15 – ALC CONFIG 1, DEFAULT 00000000 ..... 19
REGISTER 0X16 – ALC CONFIG 2, DEFAULT 00000000 ..... 19
REGISTER 0X17 – ALC LEVEL, DEFAULT 11110111 ..... 20
REGISTER 0X18 – MISC CONTROL 3, DEFAULT 00000000 ..... 21
REGISTER 0X19 – EQ CONTROL 1, DEFAULT 00100000 ..... 21
REGISTER 0X1A – EQ CONFIG 2, DEFAULT 00000000 ..... 22
REGISTER 0X20 – ANALOG SYSTEM, DEFAULT 00010110 ..... 22
REGISTER 0X21 – ANALOG SYSTEM, DEFAULT 01111111 ..... 22
REGISTER 0X22 – ANALOG SYSTEM, DEFAULT 00000000 ..... 22
REGISTER 0X23 – ANALOG SYSTEM, DEFAULT 10000110 ..... 23
REGISTER 0X24 – ANALOG SYSTEM, DEFAULT 00000000 ..... 23
REGISTER 0X25 – ANALOG SYSTEM, DEFAULT 00000111 ..... 23
REGISTER 0X40~0X5E – EQ DATA RAM CLEAR, DEFAULT 00000000 ..... 24
REGISTER 0XFC – PAGE SELECT, DEFAULT 00000000 ..... 24
REGISTER 0XFD – CHIP ID1, DEFAULT 10000001 ..... 24
REGISTER 0XFE – CHIP ID0, DEFAULT 01010101 ..... 24
REGISTER 0XFF – CHIP VERSION, DEFAULT 00010000 ..... 24
REGISTER 0X00~0XD1(PAGE 1) – EQ COEFFICENT ..... 24
*10. PACKAGE ..... 25*
*11. CORPORATE INFORMATION ..... 26*

# 1. BLOCK DIAGRAM

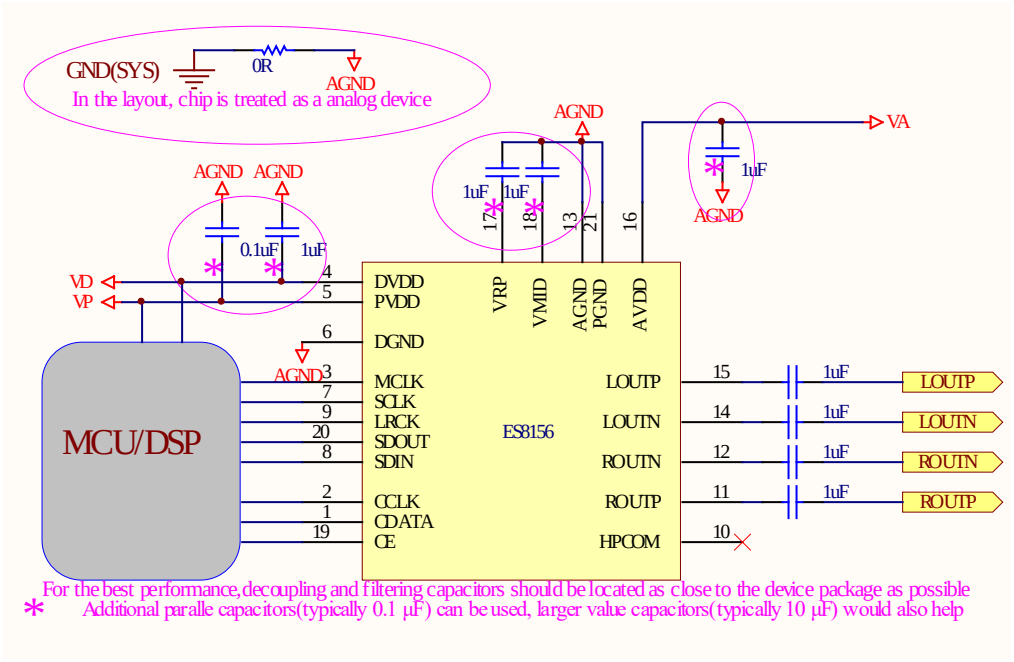


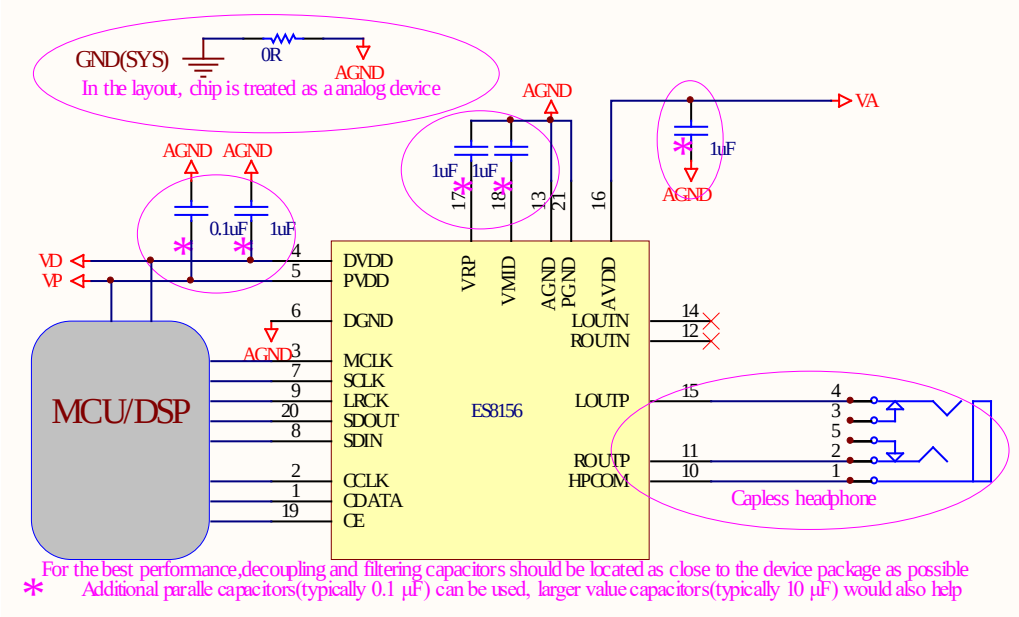
## 2. PIN OUT AND DESCRIPTION

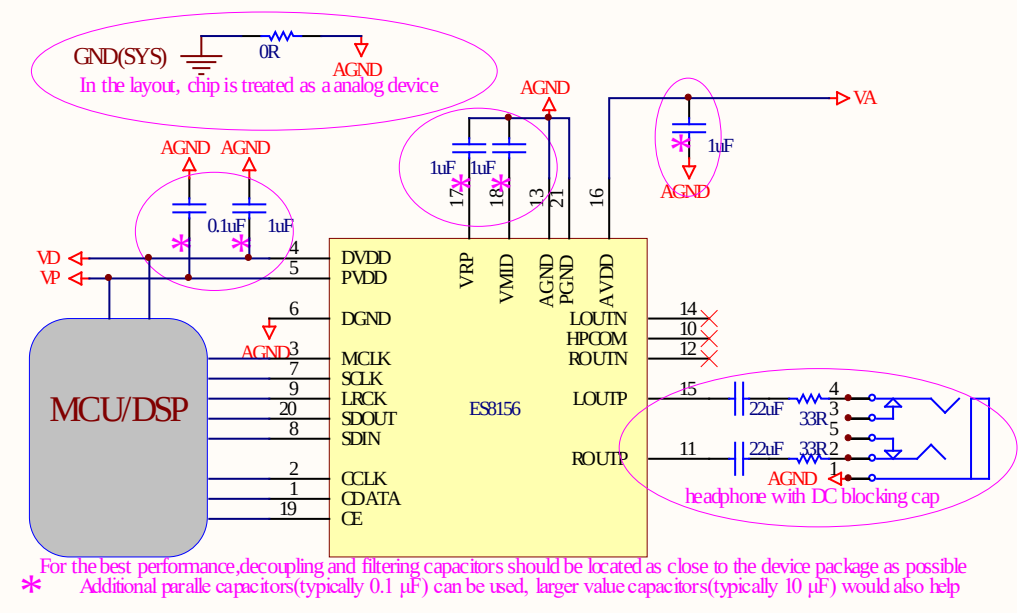



| Pin Name | Pin number | Input or Output | Pin Description |
|---|---|---|---|
| CDATA, CCLK, CE | 1, 2, 19 | I/O, I, I | $I^2C$ clock, data, address |
| MCLK | 3 | I | Master clock |
| SCLK | 7 | I/O | Serial data bit clock/DMIC bit clock |
| SDIN | 8 | I | DAC serial data input |
| LRCK | 9 | I/O | Serial data left and right channel frame clock |
| SDOUT | 20 | O | Playback signal feedback |
| LOUTP, LOUTN<br>ROUTP, ROUTN | 15, 14<br>11, 12 | O<br>O | Left channel differential analog output<br>Right channel differential analog output |
| HPCOM | 10 | Analog | Virtual ground for capless headphone<br>(Only available in software mode) |
| PVDD | 5 | Analog | Power supply for the digital input and output |
| DVDD, DGND | 4, 6 | Analog | Digital power supply |
| AVDD, AGND | 16, 13 | Analog | Analog power supply |
| VMID | 18 | Analog | Filtering capacitor connection |
| VRP | 17 | Analog | Filtering capacitor connection |

## 3. TYPICAL APPLICATION CIRCUIT




GND(SYS)
0R
AGND
In the layout, chip is treated as a analog device
AGND
VA
1uF
AGND
AGND AGND
0.1uF 1uF
1uF 1uF
17 18 13 21 16
VD
VP
4
5
6
AGND 3
7
9
20
8
2
1
19
MCU/DSP
DVDD
PVDD
DGND
MCLK
SCLK
LRCK
SDOUT
SDIN
CCLK
CDATA
CE
ES8156
VRP
VMID
AGND
PGND
AVDD
LOUTN
HPCOM
ROUTN
LOUTP
ROUTP
14
10
12
15
11
22uF
33R
4
3
5
2
1
AGND
headphone with DC blocking cap
For the best performance,decoupling and filtering capacitors should be located as close to the device package as possible
Additional paralle capacitors(typically 0.1 μF) can be used, larger value capacitors(typically 10 μF) would also help

## 4. HARDWARE MODE

The device works either in hardware mode (HW mode) or software mode ($I^2C$ mode). The default is hardware mode. Software mode is enabled by setting bit 2 of configuration register 0x02.

In HW mode, LRCK and SCLK are supplied externally. LRCK and SCLK must be synchronously derived from the system clock with some specific rates. The device can auto detect MCLK/LRCK ratio according to Table 1. The device only supports the MCLK/LRCK ratios listed in Table 1. The SCLK/LRCK ratio is normally 64.

Table 1 Slave Mode Sampling Frequencies and MCLK/LRCK Ratio

| Speed Mode | Sampling Frequency | MCLK/LRCK Ratio |
|---|---|---|
| Single Speed | 8kHz – 50kHz | 32, 64, 96, 128, 192, 256, 384, 512, 640, 768, 1024, 1152, 1280, 1536 |

## 5. CLOCK MODES AND SAMPLING FREQUENCIES

In software mode, the device supports standard audio clocks (32Fs, 64F, 128Fs, 256Fs, 384Fs, 512Fs, etc), USB clocks (12/24 MHz), and some common non standard audio clocks (16 MHz, 25 MHz, 26 MHz, etc).

According to the serial audio data sampling frequency (Fs), the device can work in two speed modes: single speed mode or double speed mode. In single speed mode, Fs normally ranges from 8 kHz to 48 kHz, and in double speed mode, Fs normally range from 64 kHz to 96 kHz.

The device can work either in master clock mode or slave clock mode. In slave mode, LRCK and SCLK are supplied externally, and LRCK and SCLK must be synchronously derived from the system clock with specific rates. In master mode, LRCK and SCLK are derived internally from device master clock.

## 6. MICRO-CONTROLLER CONFIGURATION INTERFACE

The device supports standard $I^2C$ micro-controller configuration interface. External micro-controller can completely configure the device through writing to internal configuration registers.

$I^2C$ interface is a bi-directional serial bus that uses a serial data line (CDATA) and a serial clock line (CCLK) for data transfer. The timing diagram for data transfer of this interface is given in Figure 1a and Figure 1b. Data are transmitted synchronously to CCLK clock on the CDATA line on a byte-by-byte basis. Each bit in a byte is sampled during CCLK high with MSB bit being transmitted firstly. Each transferred byte is followed by an acknowledge bit from receiver to pull the CDATA low. The transfer rate of this interface can be up to 400 kbps.

A master controller initiates the transmission by sending a "start" signal, which is defined as a high-to-low transition at CDATA while CCLK is high. The first byte transferred is the slave address. It is a seven-bit chip address followed by a RW bit. The chip address must be 0001 00x, where x equals CE. The RW bit indicates the slave data transfer direction. Once an acknowledge bit is received, the data transfer starts to proceed on a byte-by-byte basis in the direction specified by the RW bit. The master can terminate the communication by generating a "stop" signal, which is defined as a low-to-high transition at CDATA while CCLK is high.

In $I^2C$ interface mode, the registers can be written and read. The formats of "write" and "read" instructions are shown in Table 2 and Table 3. Please note that, to read data from a register, you must set R/W bit to 0 to access the register address and then set R/W to 1 to read data from the register.

Table 2 Write Data to Register in $I^2C$ Interface Mode

| | Chip Address | R/W | | Register Address | | Data to be written | | |
|---|---|---|---|---|---|---|---|---|
| start | 0001 00 CE | 0 | ACK | RAM | ACK | DATA | ACK | Stop |

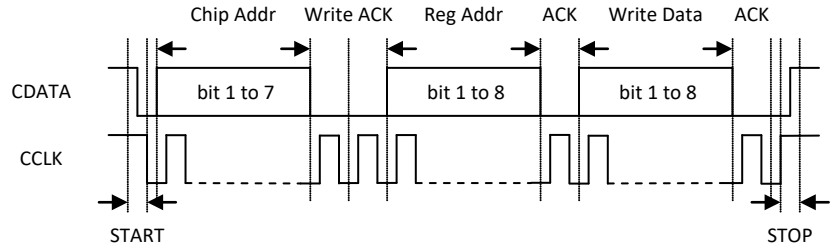



Figure 1a $I^2C$ Write Timing

Table 3 Read Data from Register in $I^2C$ Interface Mode

| | Chip Address | R/W | | Register Address | | |
|---|---|---|---|---|---|---|
| Start | 0001 00 CE | 0 | ACK | RAM | ACK | |
| | Chip Address | R/W | | Data to be read | | |
| Start | 0001 00 CE | 1 | ACK | Data | NACK | Stop |

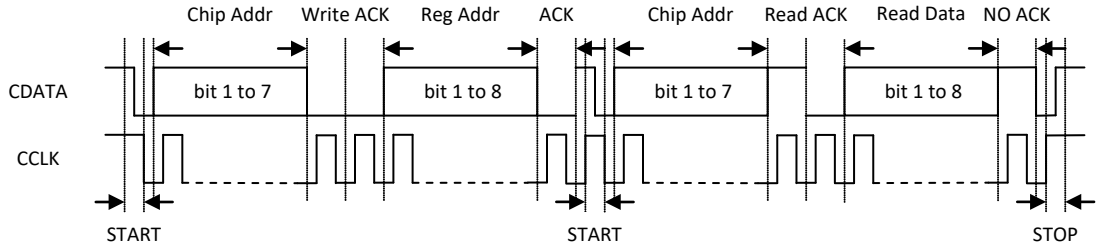



Figure 1b $I^2C$ Read Timing

## 7. DIGITAL AUDIO INTERFACE

The device provides many formats of serial audio data interface to the input or output through LRCK, SCLK and SDIN or SDOUT pins. These formats are $I^2S$, left justified, right justified and DSP/PCM. SDIN is sampled by the device on the rising edge of SCLK. SDOUT is out on the falling edge of SCLK. The relationship of SDATA (SDIN or SDOUT), SCLK and LRCK with these formats are shown through Figure 2a to Figure 2d.

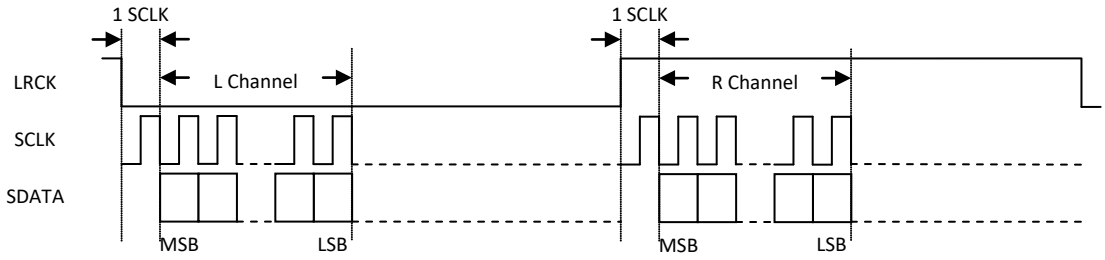



Figure 2a $I^2S$ Serial Audio Data Format

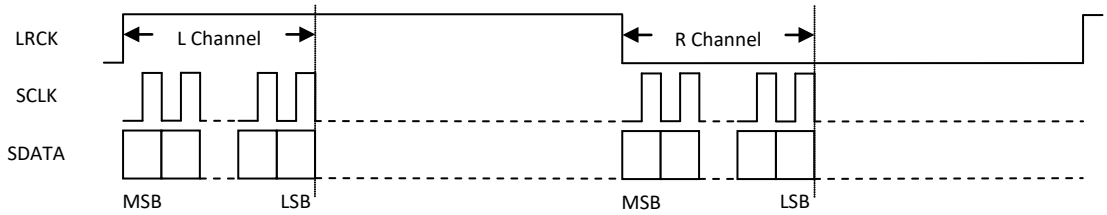



Figure 2b Left Justified Serial Audio Data Format

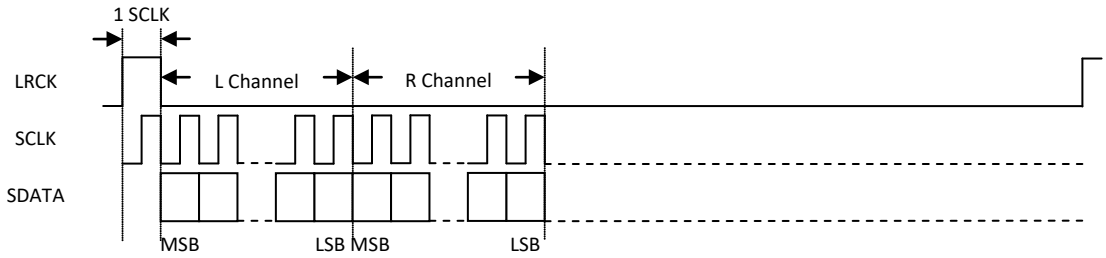



Figure 2c DSP/PCM Mode A Serial Audio Data Format

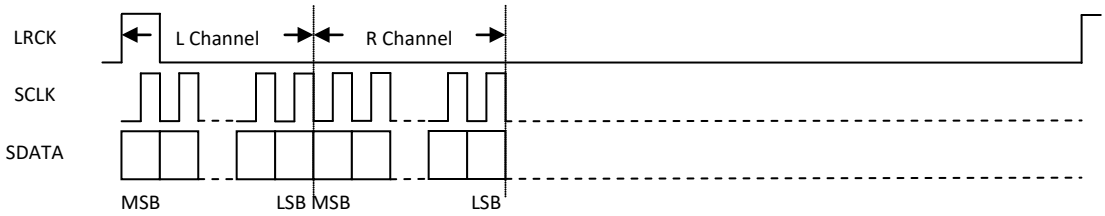



Figure 2d DSP/PCM Mode B Serial Audio Data Format

# 8. ELECTRICAL CHARACTERISTICS

***ABSOLUTE MAXIMUM RATINGS***

Continuous operation at or beyond these conditions may permanently damage the device.

| PARAMETER | MIN | MAX |
|---|---|---|
| Analog Supply Voltage Level | -0.3V | +3.6V |
| Digital Supply Voltage Level | -0.3V | +3.6V |
| Analog Input Voltage Range | AGND-0.3V | AVDD+0.3V |
| Digital Input Voltage Range | DGND-0.3V | PVDD+0.3V |
| Operating Temperature Range | -40°C | +85°C |
| Storage Temperature | -65°C | +150°C |

***RECOMMENDED OPERATING CONDITIONS***

| PARAMETER | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|
| DVDD | 1.6 | 3.3 | 3.6 | V |
| PVDD | 1.6 | 3.3 | 3.6 | V |
| AVDD | 1.7 | 3.3 | 3.6 | V |

***DAC ANALOG AND FILTER CHARACTERISTICS AND SPECIFICATIONS***

Test conditions are as the following unless otherwise specify: AVDD=3.3V, DVDD=3.3V, AGND=0V, DGND=0V, Ambient temperature=25°C, Fs=48 KHz, MCLK/LRCK=256.

| PARAMETER | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|
| DAC Performance | | | | |
| Signal to Noise ratio (A-weigh) | 105 | 110 | 115 | dB |
| THD+N | -85 | -80 | -78 | dB |
| Channel Separation (1KHz) | 105 | 110 | 115 | dB |
| Interchannel Gain Mismatch | | 0.1 | | dB |
| Filter Frequency Response – Single Speed | | | | |
| Passband | 0 | | 0.4535 | Fs |
| Stopband | 0.5465 | | | Fs |
| Passband Ripple | | | ±0.05 | dB |
| Stopband Attenuation | 53 | | | dB |
| Filter Frequency Response – Double Speed | | | | |
| Passband | 0 | | 0.4167 | Fs |
| Stopband | 0.7917 | | | Fs |
| Passband Ripple | | | ±0.005 | dB |
| Stopband Attenuation | 56 | | | dB |
| Analog Output | | | | |
| Full Scale Output Level | | AVDD/3.3 | | Vrms |

***DC CHARACTERISTICS***

| PARAMETER | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|
| Normal Operation Mode | | | | |
| DVDD=1.8V, PVDD=1.8V, AVDD=3.3V<br>DVDD=1.8V, PVDD=1.8V, AVDD=1.8V | | 19<br>6 | | mW |
| Power Down Mode | | | | |
| DVDD=1.8V, PVDD=1.8V, AVDD=3.3V | | 0 | | uA |
| Digital Voltage Level | | | | |
| Input High-level Voltage | 0.7*PVDD | | | V |
| Input Low-level Voltage | | | 0.5 | V |
| Output High-level Voltage | | PVDD | | V |
| Output Low-level Voltage | | 0 | | V |

***$I^2C$ SWITCHING SPECIFICATIONS (SLOW SPEED MODE/HIGH SPEED MODE)***

| PARAMETER | Symbol | MIN | MAX | UNIT |
|---|---|---|---|---|
| CCLK Clock Frequency | $F_{CCLK}$ | | 100/400 | KHz |
| Bus Free Time Between Transmissions | $T_{TWID}$ | 4.7/1.3 | | us |
| Start Condition Hold Time | $T_{TWSTH}$ | 4.0/0.6 | | us |
| Clock Low time | $T_{TWCL}$ | 4.7/1.3 | | us |
| Clock High Time | $T_{TWCH}$ | 4.0/0.6 | | us |
| Setup Time for Repeated Start Condition | $T_{TWSTS}$ | 4.7/0.6 | | us |
| CDATA Hold Time from CCLK Falling | $T_{TWDH}$ | | 3.45/0.9 | us |
| CDATA Setup time to CCLK Rising | $T_{TWDS}$ | 0.25/0.1 | | us |
| Rise Time of CCLK | $T_{TWR}$ | | 1.0/0.3 | us |
| Fall Time CCLK | $T_{TWF}$ | | 1.0/0.3 | us |

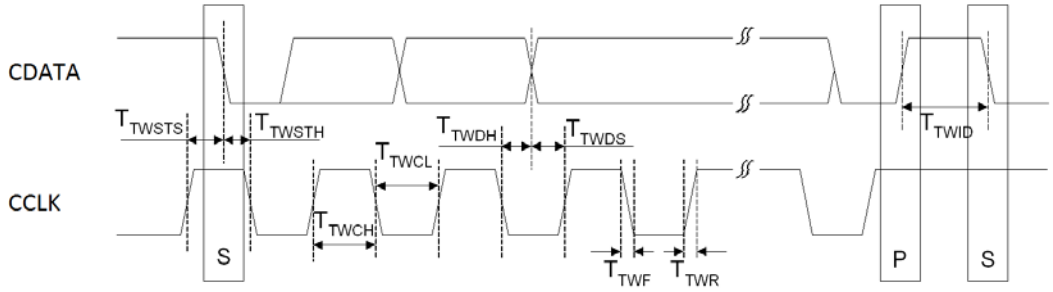



Figure 3 $I^2C$ Timing

***SERIAL AUDIO PORT SWITCHING SPECIFICATIONS***

| PARAMETER | Symbol | MIN | MAX | UNIT |
|---|---|---|---|---|
| MCLK frequency | | | 49.2 | MHz |
| MCLK duty cycle | | 40 | 60 | % |
| LRCK frequency | | | 200 | KHz |
| LRCK duty cycle (Note 2) | | 40 | 60 | % |
| SCLK frequency | | | 26 | MHz |
| SCLK pulse width low | $T_{SLKL}$ | 16 | | ns |
| SCLK Pulse width high | $T_{SCLKH}$ | 16 | | ns |
| SCLK falling to LRCK edge (master mode only) | $T_{SLR}$ | | 10 | ns |
| LRCK edge to SCLK rising (slave mode only) | $T_{LSR}$ | 10 | | ns |
| SCLK falling to SDOUT valid VDDD=3.3V<br>VDDD=1.8V | $T_{SDO}$ | | 16<br>39 | ns |
| LRCK edge to SDOUT valid (Note 3) VDDD=3.3V<br>VDDD=1.8V | $T_{LDO}$ | | 11<br>25 | ns |
| SDIN valid to SCLK rising setup time | $T_{SDIS}$ | 10 | | ns |
| SCLK rising to SDIN hold time | $T_{SDIH}$ | 10 | | ns |

Note 2: one SCLK period of high time in DSP/PCM modes.

Note 3: only apply to MSB of Left Justified or DSP/PCM mode B.

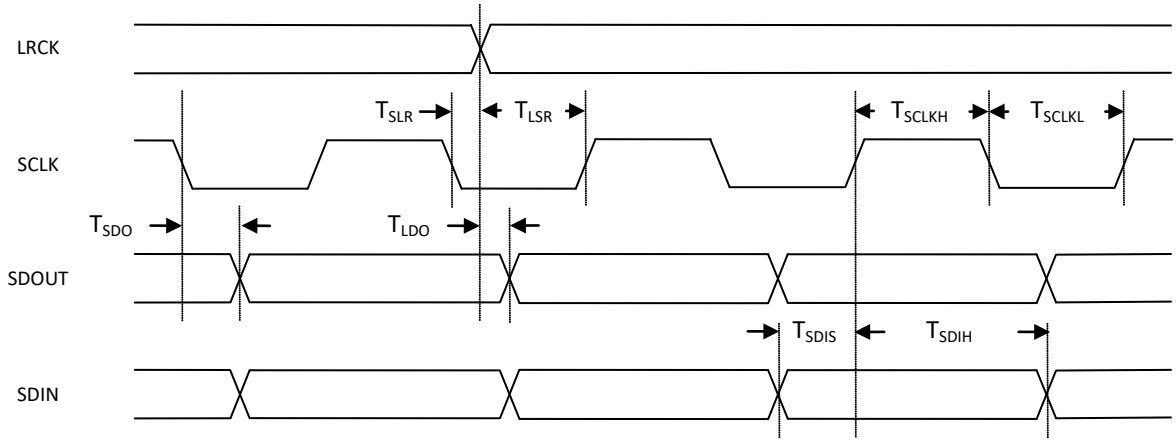



Figure 4 Serial Audio Port Timing

# 9. CONFIGURATION REGISTER DEFINITION

Registers reside in two pages. Page 1 is dedicated to EQ coefficients. All other registers are in page 0. Page select is in register 0xFC.

***REGISTER 0X00 – RESET CONTROL, DEFAULT 00011100***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:6 | Reserved |
| RST_REGS | 5 | 0 – normal<br>1 – reset control registers |
| RST_MSTGEN | 4 | 0 – normal<br>1 – reset master mode LRCK and SCLK |
| RST_DAC_DIG | 3 | 0 – normal<br>1 – reset DAC |
| RST_DIG | 2 | 0 – normal<br>1 – reset digital |
| SEQ_DIS | 1 | Auto power sequence<br>0 – enable<br>1 – disable |
| CSM_ON | 0 | Chip state machine power down<br>0 – disable<br>1 – enable |

***REGISTER 0X01 – MAIN CLOCK CONTROL, DEFAULT 00100000***

| Bit Name | Bit | Description |
|---|---|---|
| MULTP_FACTOR | 7:6 | Master clock multiplier<br>00 – multiply by 1<br>01 – multiply by 2<br>10 – multiply by 4<br>11 – multiply by 8 |
| OSR128_SEL | 5 | 1 – 128 OSR<br>0 – 64 OSR |
| CLK_DAC_DIV | 4:0 | DAC clock divide<br>0/1 – no divide<br>2 – divide by 2<br>…..<br>31 – divide by 31 |

***REGISTER 0X02 – MODE CONFIG 1, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| SCLK_AS_MCLK | 7 | 0 – MCLK is as MCLK<br>1 – SCLK is as MCLK |
| ISCLKLRCK_SEL | 6 | 0 – according to M/S mode<br>1 – master mode's LRCK/SCLK as internal LRCK/SCLK for speedy start up |
| SCLKLRCK_TRI | 5 | 0 – original data<br>1 – SCLK/LRCK is tri state output |
| SCLK_INV_MODE | 4 | SCLK invert<br>0 – normal<br>1 – SCLK invert |

| | | |
|---|---|---|
| EQ_HIGH_MODE | 3 | 0 – 1x frequency<br>1 – 2x frequency |
| SOFT_MODE_SEL | 2 | software mode/hardware mode select<br>0 – hardware mode<br>1 – software mode |
| SPEED_MODE | 1 | 0 – single speed<br>1 – double speed |
| MS_MODE | 0 | LRCK/SCLK clock mode<br>0 – slave mode<br>1 – master mode |

***REGISTER 0X03 – MASTER LRCK DIVIDER 1, DEFAULT 00000001***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:4 | Reserved |
| M_LRCK_DIV_HIGH4 | 3:0 | M_LRCK_DIV[11:8] |

***REGISTER 0X04 – MASTER LRCK DIVIDER 0, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| M_LRCK_DIV_LOW8 | 7:0 | M_LRCK_DIV[7:0] |

***REGISTER 0X05 – MASTER CLOCK CONTROL, DEFAULT 00000100***

| Bit Name | Bit | Description |
|---|---|---|
| M_SCLK_MODE | 7 | 0 – normal<br>1 – reserved |
| M_SCLK_DIV | 6:0 | SCLK divide (use with MSTCLK_SRCSEL)<br>0/1 – no divide<br>2 – divide by 2<br>....<br>127 – divide by 127 |

***REGISTER 0X06 – NFS CONFIG, DEFAULT 00010001***

| Bit Name | Bit | Description |
|---|---|---|
| LRCK_CNT_RV | 7:4 | Use for debug, Read Only |
| LRCK_RATE_MODE | 3:0 | 1 – 2 channels, normal mode<br>2 – 4 channels<br>3 – 6 channels<br>4 – 8 channels<br>5 – 10 channels<br>6 – 12 channels<br>7 – 14 channels<br>8 – 16 channels<br>Other – reserved |

***REGISTER 0X07 – MISC CONTROL 1, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| MCLK_INV | 7 | 0 – normal<br>1 – invert MCLK |
| CLK_DAC_DIV0 | 6:4 | DAC clock divide<br>0/1 – no divide<br>2 – divide by 2 |

| | | |
|---|---|---|
| | | …..<br>7 – divide by 7 |
| CLKDBL_PW_SEL | 3:2 | 0/1 – normal clock doubler duty cycle<br>2 – clock doubler duty cycle1<br>3 – clock doubler duty cycle2 |
| CLKDBL_PATH_SEL | 1 | Clock doubler path select (internal use) |
| LRCK_EXTEND | 0 | 0 – normal internal LRCK pulse width<br>1 – extended internal LRCK pulse width |

***REGISTER 0X08 – CLOCK OFF, DEFAULT 00000110***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:6 | Reserved |
| P2S_CLK_ON | 5 | 0 – turn off P2S master clock<br>1 – turn on P2S master clock |
| MASTER_CLK_ON | 4 | 0 – turn off master mode SCLK and LRCK<br>1 – turn on master mode SCLK and LRCK |
| EXT_SCLKLRCK_ON | 3 | 0 – turn off slave mode SCLK and LRCK<br>1 – turn on slave mode SCLK and LRCK |
| ANA_CLK_ON | 2 | 0 – turn off DAC analog clock<br>1 – turn on DAC analog clock |
| DAC_MCLK_ON | 1 | 0 – turn off DAC master clock<br>1 – turn on DAC master clock |
| MCLK_ON | 0 | 0 – turn off master clock<br>1 – turn on master clock |

***REGISTER 0X09 – MISC CONTROL 2, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:6 | Reserved |
| MSTCLK_SRCSEL | 5 | 0 – source clock of master mode from pad<br>1 – source clock of master mode from internal clock |
| CSM_CNTSEL | 4 | 0 – slave LRCK used for CSM counter<br>1 – master LRCK used for CSM counter |
| Reserved | 3:2 | Reserved |
| DLL_ON | 1 | 1 – normal<br>0 – power down DLL |
| PUPDN_OFF | 0 | Pin internal pull up<br>1 – off<br>0 – on |

***REGISTER 0X0A – TIME CONTROL 1, DEFAULT 01010000***

| Bit Name | Bit | Description |
|---|---|---|
| V_T1 | 7:0 | Timer1<br>For LRCK=48KHz:<br>0 - NA<br>1 - 80us(default)<br>2 – 2.5ms<br>…<br>63 - 675ms(max) |

***REGISTER 0X0B – TIME CONTROL 2, DEFAULT 01010000***

| Bit Name | Bit | Description |
|---|---|---|
| V_T2 | 7:0 | Timer2<br>For LRCK=48KHz:<br>0 - NA<br>1 - 80us(default)<br>2 – 2.5ms<br>…<br>63 - 675ms(max) |

***REGISTER 0X0C – CHIP STATUS, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7 | Reserved |
| CSM_STATE | 6:4 | Chip state machine state, Read Only<br>000 – S0<br>001 – S1<br>010 – S2<br>011 – S3<br>110 – S6<br>Other – NA |
| Reserved | 3 | Reserved |
| FORCE_CSM | 2:0 | Force chip state machine<br>100 – force to S0<br>101 – force to S1<br>110 – force to S2<br>111 – force to S3<br>110 – force to S6<br>Other – no force |

***REGISTER 0X0D – P2S CONTRL, DEFAULT 00010000***

| Bit Name | Bit | Description |
|---|---|---|
| LRCK_1STCNT | 7:4 | Reserved |
| P2S_NFS_FLAGOFF | 3 | 0 – include flag in P2S data<br>1 – don't include flag in P2S data |
| P2S_SDOUT_MUTEB | 2 | 0 – mute P2S output<br>1 – output P2S data |
| P2S_SDOUT_SEL | 1 | 0 – normal<br>1 – only use under test mode |
| P2S_SDOUT_TRI | 0 | SDOUT tri-state control<br>0 – normal output<br>1 – high z output |

***REGISTER 0X10 – DAC COUNTER PARAMETER, DEFAULT 01000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7 | Reserved |
| DAC_NS | 6:0 | OSR parameter selection<br>>=64 |

***REGISTER 0X11 – SDP INTERFACE CONFIG 1, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|

| | | |
|---|---|---|
| Reserved | 7 | Reserved |
| SP_WL | 6:4 | 000 – 24-bit<br>001 – 20-bit<br>010 – 18-bit<br>011 – 16-bit<br>100 – 32-bit |
| SP_MUTE | 3 | Software mute control<br>0 – normal<br>1 – mute DAC input data to 0 |
| SP_LRP | 2 | $I^2S$/Left Justified:<br>0 – L/R normal polarity<br>Left/Right=High/Low (Left Justified)<br>Left/Right=Low/High ($I^2S$)<br>1 – L/R invert polarity<br>Left/Right=Low/High (Left Justified)<br>Left/Right=High/Low ($I^2S$)<br>DSP/PCM mode:<br>0 – mode A<br>1 – mode B |
| SP_PROTOCAL | 1:0 | 00 – $I^2S$<br>01 – Left Justified<br>10 – Reserved<br>11 – DSP mode |

***REGISTER 0X12 – AUTOMUTE CONTROL, DEFAULT 00000100***

| Bit Name | Bit | Description |
|---|---|---|
| AUTOMUTE_NG | 7:4 | DAC auto mute noise gate<br>0000 – -96dB<br>0001 – -90dB<br>0010 – -84dB<br>0011 – -78dB<br>0100 – -72dB<br>0101 – -66dB<br>0110 – -60dB<br>0111 – -54dB<br>1000 – -51dB<br>1001 – -48dB<br>1010 – -45dB<br>1011 – -42dB<br>1100 – -39dB<br>1101 – -36dB<br>1110 – -33dB<br>1111 – -30dB |
| AUTOMUTE_SIZE | 3:0 | Auto mute window size<br>0 – 2048<br>other – 2048 * AUTOMUTE_SIZE |

***REGISTER 0X13 – MUTE CONTROL, DEFAULT 00010001***

| Bit Name | Bit | Description |
|---|---|---|
| INTOUT_CLIPEN | 7 | DSP clip out control<br>0 – normal |

| | | |
|---|---|---|
| | | 1 – clip out |
| AM_ATTENU6_ENA | 6 | DSP clip out control<br>0 – normal<br>1 – enable attenuation |
| AM_ACLKOFF_ENA | 5 | 0 – normal<br>1 – disable analog clock |
| AM_DSMMUTE_ENA | 4 | 0 – normal<br>1 – mute DSM output |
| AM_MUTE_FLAG | 3 | Reserved |
| LCH_DSM_SMUTE | 2 | 0 – normal<br>1 – mute DSM output of left channel |
| RCH_DSM_SMUTE | 1 | 0 – normal<br>1 – mute DSM output of right channel |
| AM_ENA | 0 | 0 – automute off<br>1 – automute on |

***REGISTER 0X14 – VOLUME CONTROL, DEFAULT 10111111***

| Bit Name | Bit | Description |
|---|---|---|
| DAC_VOLUME_DB | 7:0 | 0x00 - '-95.5dB' (default)<br>0x01 - '-95.0dB'<br>⋯ 0.5dB/step<br>0xBE - '-0.5dB'<br>0xBF - '0dB'<br>0xC0 - '+0.5dB'<br>⋯<br>0xFF - '+32dB' |

***REGISTER 0X15 – ALC CONFIG 1, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:4 | Reserved |
| ALC_MUTE_GAIN | 3:1 | ALC gain control when automute<br>000 – 0dB<br>001 – -4dB<br>010 – -8dB<br>011 – -12dB<br>100 – -16dB<br>101 – -20dB<br>110 – -24dB<br>111 – -28dB |
| DAC_ALC_EN | 0 | 0 – ALC off<br>1 – ALC on |

***REGISTER 0X16 – ALC CONFIG 2, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| ALC_RAMP_RATE | 7:4 | DAC ALC ramp rate<br>For LRCK=48KHz:<br>1 - 0.25dB/80us<br>2 - 0.25dB /160us<br>3 - 0.25dB /320us<br>4 - 0.25dB /640us |

| | | |
|---|---|---|
| | | 5 - 0.25dB /1.28ms<br>6 - 0.25dB /2.56ms<br>7 - 0.25dB /5.12ms<br>8 - 0.25dB /10.24ms<br>9 - 0.25dB /20.48ms<br>10 - 0.25dB /40.96ms<br>11 - 0.25dB /81.92ms<br>12 - 0.25dB /163.84ms<br>13 - 0.25dB /327.68ms<br>14 - 0.25dB /655.36ms<br>15 - 0.25dB /1.3s |
| ALC_WIN_SIZE | 3:0 | DAC ALC window size<br>For LRCK=48KHz:<br>1 - 0.25dB/40us<br>2 - 0.25dB /80us<br>3 - 0.25dB /160us<br>4 - 0.25dB /320us<br>5 - 0.25dB /640us<br>6 - 0.25dB / 1.28ms<br>7 - 0.25dB / 2.56ms<br>8 - 0.25dB /5.12ms<br>9 - 0.25dB /10.24ms<br>10 - 0.25dB /20.48ms<br>11 - 0.25dB /40.96ms<br>12 - 0.25dB /81.92ms<br>13 - 0.25dB /163.84ms<br>14 - 0.25dB /327.68ms<br>15 - 0.25dB /655.36ms |

***REGISTER 0X17 – ALC LEVEL, DEFAULT 11110111***

| Bit Name | Bit | Description |
|---|---|---|
| ALC_MAXLEVEL | 7:4 | ALC max level<br>15 – -6.0 dB<br>14 – -6.6 dB<br>13 – -7.2 dB<br>12 – -7.8 dB<br>11 – -8.5 dB<br>10 – -9.3 dB<br>9 – -10.1dB<br>8 – -11.0dB<br>7 – -12.0dB<br>6 – -13.2dB<br>5 – -14.5dB<br>4 – -16.1dB<br>3 – -18.1dB<br>2 – -20.6dB<br>1 – -24.1dB<br>0 – -30.1dB |
| ALC_MINLEVEL | 3:0 | ALC min level<br>15 – -6.0 dB<br>14 – -6.6 dB<br>13 – -7.2 dB |

| | | |
|---|---|---|
| | | 12 – -7.8 dB<br>11 – -8.5 dB<br>10 – -9.3 dB<br>9 – -10.1dB<br>8 – -11.0dB<br>7 – -12.0dB<br>6 – -13.2dB<br>5 – -14.5dB<br>4 – -16.1dB<br>3 – -18.1dB<br>2 – -20.6dB<br>1 – -24.1dB<br>0 – -30.1dB |

***REGISTER 0X18 – MISC CONTROL 3, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| P2S_DATA_BITNUM | 7 | DSP clip out control<br>0 – 24 bit<br>1 – 26 bit |
| P2S_DPATH_SEL | 6 | DSP clip out control<br>0 – EQ output<br>1 – DSP output |
| CHN_CROSS | 5:4 | 00 – normal<br>01 – left channel to both left and right channels<br>10 – right channel to both left and right channels<br>11 – left and right channel output switch |
| LCH_INV | 3 | 0 – normal<br>1 – invert left channel output |
| RCH_INV | 2 | 0 – normal<br>1 – invert right channel output |
| DSM_DITHERON | 1 | 0 – normal<br>1 – open dither of DSM |
| DAC_RAM_CLR | 0 | 0 – normal<br>1 – clear DSP ram |

***REGISTER 0X19 – EQ CONTROL 1, DEFAULT 00100000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7 | Reserved |
| EQ_BAND_NUM | 6:4 | 000 – 1 left and right band<br>001 – 2 left and right band<br>010 – 3 left and right band<br>(3.3V DVDD required for 48 kHz Fs)<br>011 – 4 left and right band<br>100 – 5 left and right band (3.3V DVDD for 48 kHz Fs)<br>101 – 6 left and right band (3.3V DVDD for 48 kHz Fs)<br>110 – 7 left and right band (3.3V DVDD for 48 kHz Fs)<br>111 – NA |
| EQ_RST | 3 | 0 – normal<br>1 – reset EQ logic |
| EQ_CFG_RD | 2 | 0 – EQ RAM read controlled by internal logic<br>1 – EQ RAM read controlled by $I^2C$ |

| | | |
|---|---|---|
| EQ_CFG_WR | 1 | 0 – EQ RAM write controlled by internal logic<br>1 – EQ RAM write controlled by $I^2C$ |
| EQ_ON | 0 | 0 – EQ off<br>1 – EQ on |

***REGISTER 0X1A – EQ CONFIG 2, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| EQ_1STCNT_VLD | 7 | 0 – the parameter of EQ_RUN_1STCNT is invalid<br>1 – the parameter of EQ_RUN_1STCNT is valid |
| EQ_RUN_1STCNT | 6:0 | Internal |

***REGISTER 0X20 – ANALOG SYSTEM, DEFAULT 00010110***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:6 | Reserved |
| S3_SEL | 5:4 | SEL S3<br>00 – NA<br>01 – default<br>10 – state2<br>11 – state3 |
| S2_SEL | 3:2 | SEL S2<br>00 – NA<br>01 – default<br>10 – state2<br>11 – state3 |
| S6_SEL | 1:0 | SEL S6<br>00 – NA<br>01 – default<br>10 – state2<br>11 – state3 |

***REGISTER 0X21 – ANALOG SYSTEM, DEFAULT 01111111***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7 | Reserved |
| VREF_RMPDN2 | 6 | 0 – normal<br>1 – fast option |
| VREF_RMPDN1 | 5 | 0 – normal<br>1 – slow option |
| VSEL | 4:0 | Select bias<br>001 - 25%<br>010 - 50%<br>011 - 75%<br>100 – 100%<br>101 - 120%<br>110 - 150%<br>111 - 175% |

***REGISTER 0X22 – ANALOG SYSTEM, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:4 | Reserved |
| HPSW | 3 | 0 – line out |

| | | |
|---|---|---|
| | | 1 – enable HP driver for headphone output |
| Reserved | 2 | |
| SWHWSEL | 1 | 0 – hardware mode<br>1 – software mode |
| OUT_MUTE | 0 | 0 – normal<br>1 – mute output |

***REGISTER 0X23 – ANALOG SYSTEM, DEFAULT 10000110***

| Bit Name | Bit | Description |
|---|---|---|
| IBIAS_SW | 7:6 | 0 –bias setting level0（lowest）<br>1 –bias setting level1<br>2 –bias setting level2<br>3 –bias setting level3 |
| VMIDLVL | 5:4 | 0 –'vdda/2' (default)<br>1 – vmid level1<br>2 – vmid level2<br>3 – vmid level3 |
| DAC_IBIAS_SW | 3 | 0 – normal DAC bias setting<br>1 – higher DAC bias setting |
| VROI | 2 | 0 – normal impedance<br>1 – low impedance |
| HPCOM_REF2 | 1 | 0 – normal HPCOM output bias<br>1 – high level HPCOM bias |
| HPCOM_REF1 | 0 | 0 – normal output bias<br>1 – low level HPCOM bias |

***REGISTER 0X24 – ANALOG SYSTEM, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:4 | Reserved |
| LPDAC | 3 | 0 – normal mode<br>1 – low power mode for DAC |
| LPDACVRP | 2 | 0 – normal mode<br>1 – low power mode for DAC reference |
| LPHPCOM | 1 | 0 – normal mode<br>1 – low power mode for HP output reference |
| LPVREFBUF | 0 | 0 – normal mode<br>1 – low power mode for internal reference |

***REGISTER 0X25 – ANALOG SYSTEM, DEFAULT 00000111***

| Bit Name | Bit | Description |
|---|---|---|
| PDN_ANA | 7 | 0 – enable analog circuits<br>1 – power down analog circuits |
| ENREFR | 6 | 0 – disable reference circuits for output<br>1 – enable reference circuits for DAC output |
| VMIDSEL | 5:4 | 0 – VMID power down<br>1 – VMID speed charge1<br>2 – normal VMID operation<br>3 – VMID speed charge3 |
| ENHPCOM | 3 | 0 – disable HPCOM<br>1 – enable HPCOM |

| | | |
|---|---|---|
| PDN_DACVREFGEN | 2 | 0 – enable analog DAC reference circuits<br>1 – power down analog DAC reference circuits |
| PDN_VREFBUF | 1 | 0 – disable internal reference circuits<br>1 – enable reference circuits |
| PDN_DAC | 0 | 0 – enable DAC<br>1 – power down DAC |

***REGISTER 0X40~0X5E – EQ DATA RAM CLEAR, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| EQ_RAM_WDATA | 7:0 | Software write 0 data during these address space. |

***REGISTER 0XFC – PAGE SELECT, DEFAULT 00000000***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:1 | Reserved |
| PAGE_SEL | 0 | 0 – $I^2C$ work at the address space of page 0<br>1 – $I^2C$ work at the address space of page 1 |

***REGISTER 0XFD – CHIP ID1, DEFAULT 10000001***

| Bit Name | Bit | Description |
|---|---|---|
| Chip_number_id1 | 7:0 | Chip ID |

***REGISTER 0XFE – CHIP ID0, DEFAULT 01010101***

| Bit Name | Bit | Description |
|---|---|---|
| Chip_number_id0 | 7:0 | Chip ID |

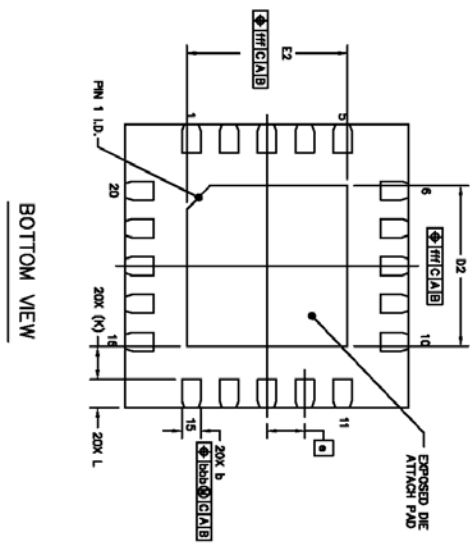
***REGISTER 0XFF – CHIP VERSION, DEFAULT 00010000***

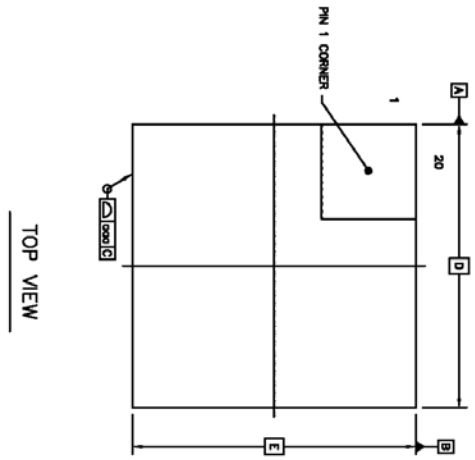
| Bit Name | Bit | Description |
|---|---|---|
| Chip_version_id1 | 7:4 | Version ID |
| Chip_version_id0 | 3:0 | Version ID |

***REGISTER 0X00~0XD1(PAGE 1) – EQ COEFFICENT***

| Bit Name | Bit | Description |
|---|---|---|
| Reserved | 7:5 | Reserved |
| EQ_COEF | 4:0 | EQ coefficient |

## 10.PACKAGE







| | SYMBOL | MIN | NOM | MAX |
|---|---|---|---|---|
| TOTAL THICKNESS | A | 0.5 | 0.55 | 0.6 |
| STAND OFF | A1 | 0 | 0.02 | 0.05 |
| MOLD THICKNESS | A2 | ---- | 0.4 | ---- |
| L/F THICKNESS | A3 | | 0.152 REF | |
| LEAD WIDTH | b | 0.15 | 0.2 | 0.25 |
| BODY SIZE X | D | | 3 BSC | |
| BODY SIZE Y | E | | 3 BSC | |
| LEAD PITCH | e | | 0.4 BSC | |
| EP SIZE X | D2 | 1.6 | 1.7 | 1.8 |
| EP SIZE Y | E2 | 1.6 | 1.7 | 1.8 |
| LEAD LENGTH | L | 0.2 | 0.3 | 0.4 |
| LEAD TIP TO EXPOSED PAD EDGE | K | | 0.35 REF | |
| PACKAGE EDGE TOLERANCE | aaa | | 0.1 | |
| MOLD FLATNESS | ccc | | 0.1 | |
| COPLANARITY | eee | | 0.08 | |
| LEAD OFFSET | bbb | | 0.1 | |
| EXPOSED PAD OFFSET | fff | | 0.1 | |

## 11.CORPORATE INFORMATION

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